

Question # 10 of 10 (Start time: 10:32:31 PM, 06 December 2023)

Total M

Which is the formula of separate type ratio estimator (using 2 strata)?

Select the correct option

 Reload Math Equation

$$\bar{y}_{rsst} = \sum_{h=1}^2 W_h \bar{y}_{rh}$$



$$\bar{y}_{rsst} = \sum_{h=1}^5 W_h \bar{y}_{rh}$$



$$\bar{y}_{rsst} = \sum_{h=1}^4 W_h \bar{y}_{rh}$$



$$\bar{y}_{rsst} = \sum_{h=1}^3 W_h \bar{y}_{rh}$$

Question # 7 of 10 (Start time: 10:30:47 PM, 06 December 2023)

Total Marks: 1

What is the R command for writing variance-covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.65 \\ 0.65 & 1 \end{bmatrix}$$

?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☐ | None of the above |
| ☒ | <code>sig=matrix(c(1,0.65,0.65,1),ncol=2)</code> |
| ☐ | <code>sig=matrix(c(1,1,0.65,0.65),ncol=2);</code> |
| ☐ | <code>sig=matrix(c(1,0.65,0.65,1),ncol=4)</code> |

Question # 5 of 10 (Start time: 10:29:45 PM, 06 December 2023)

Total Marks: 1

If $N = 6000$, $N_1 = 4000$; $n_1 = 150$, $N_2 = 2000$; $n_1 = 15$; $n_2 = 15$, the value of W_2 is:

Select the correct option

- | | |
|----------------------------------|------|
| ☒ | 0.33 |
| ☐ | 0.43 |
| ☐ | 0.38 |
| ☐ | 0.67 |

Question # 4 of 10 (Start time: 10:28:45 PM, 06 December 2023)

Total Marks: 1

The expression of Bias of product estimator in double sampling is:

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☐ | $Bias(\bar{y}_{pd}) = \bar{Y}[(\lambda_2 - \lambda_1)C_y C_x]$ |
| ☐ | $Bias(\bar{y}_{pd}) = \bar{Y}[(\lambda_2 - \lambda_1)\rho C_y C_x]$ |
| ☐ | $Bias(\bar{y}_{pd}) = \bar{Y}[(\lambda_1 - \lambda_2)\rho C_y C_x]$ |
| ☒ | $Bias(\bar{y}_{pd}) = \bar{Y}[(\lambda_2 - \lambda_1)\rho C_y C_x]$ |



Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 10:28:17 PM, 06 December 2023)

Total Marks: 1

Which of the following would usually require the smallest sample size because of its efficiency?

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | Two stage cluster sampling |
| ☐ | Quota sampling |
| ☒ | Simple random sampling |
| ☐ | One stage cluster sampling |

What is the formula of ratio estimator?

Select the correct option



None of the above



$$\frac{\bar{y}}{\bar{x}} \bar{X}$$



$$\frac{\bar{x}}{\bar{y}} \bar{X}$$



$$\frac{\bar{x}}{\bar{y}} \bar{Y}$$

Farmer Joe randomly picks 100 trees using a random number generator to estimate the number of apples produced by his apple trees. This is ----- sampling

Select the correct option

☐	Systematic
☐	Cluster
☐	Stratified
☐	Simple

Question # 8 of 10 (Start time: 10:22:53 PM, 06 December 2023)

Total Mark

Farmer Joe separates his farm into 10 regions. He then randomly selected 5 trees from each region to estimate the number of apples produced on his apple tree farm. This is ----- sampling.

Select the correct option

☐	Systematic
☐	Simple
☐	Stratified
☐	Cluster

Click to Save Answer & Move to Next Question

Question # 7 of 10 (**Start time: 10:21:53 PM, 06 December 2023**)

Which is the formula of ratio estimator in double sampling?

Select the correct option

☒	 $\bar{y}_{rd} = \frac{\bar{y}_2}{\bar{x}_2} \bar{x}_1$
☐	$\bar{y}_{rd} = \frac{\bar{y}_1}{\bar{x}_2} \bar{x}_1$
☐	None of the above
☐	$\bar{y}_{rd} = \frac{\bar{y}_2}{\bar{x}_1} \bar{x}_2$

Which is the formula of separate type ratio estimator (using 4 strata)?

Select the correct option



$$\bar{y}_{rsst} = \sum_{h=1}^4 W_h \bar{y}_{rh}$$



$$\bar{y}_{rsst} = \sum_{h=1}^4 W_h \bar{y}_{rh}$$



$$\bar{y}_{rsst} = \sum_{h=1}^4 W_h \bar{y}_{rh}$$



$$\bar{y}_{rsst} = \sum_{h=1}^4 W_h \bar{y}_{rh}$$

What is the R command for writing variance-covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.75 \\ 0.75 & 1 \end{bmatrix}$$

?

Select the correct option

☐	None of the above
☐	<code>sig=matrix(c(1,1,0.75,0.75),ncol=2);</code>
☐	<code>sig=matrix(c(1,0.75,0.75,1),ncol=4)</code>
☐	<code>sig=matrix(c(1,0.75,0.75,1),ncol=2)</code> ✓

To develop interval estimate of any parameter of population, value which is added or subtracted from point estimate is classified as:

Select the correct option

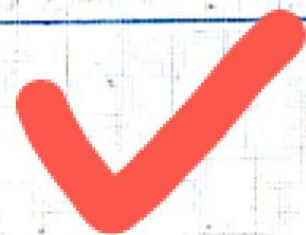
☐	margin of consistency
☐	margin of biasedness
☐	margin of error
☐	margin of efficiency



Which of the following would usually require the smallest sample size because of its efficiency?

Select the correct option

☐	Two stage cluster sampling
☐	One stage cluster sampling
☐	Quota sampling
☐	Simple random sampling



What is the R command for writing variance-covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.95 \\ 0.95 & 1 \end{bmatrix}$$

?

Select the correct option



`sig=matrix(c(1,1,0.65,0.65),ncol=2);`



`sig=matrix(c(1,0.95,0.95,1),ncol=2)`



None of the above



`sig=matrix(c(1,0.95,0.95,1),ncol=4)`

Question # 2 of 10 (Start time: 10:51:21 PM, 06 December 2023)

What is MSE of product estimator in double sampling?

Select the correct option



None of the above



$$MSE(\bar{y}_{rd}) = \bar{X}^2 [\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 + 2C_x C_y \rho_{xy})]$$



$$MSE(\bar{y}_{rd}) = \bar{Y}^2 [\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 + 2C_x C_y \rho_{xy})]$$

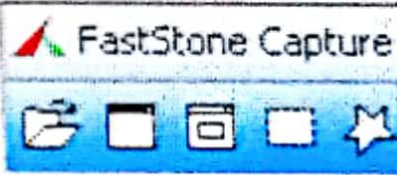


$$MSE(\bar{y}_{rd}) = \bar{Y}^2 [\lambda_2 C_x^2 + (\lambda_2 - \lambda_1)(C_y^2 + 2C_x C_y \rho_{xy})]$$

If $N= 4000$, $N1=2000$; $n1=150$, $N2=1500$; $N3 =500$; $n2=8$; $n3=25$, the value of $W3$ is:

Select the correct option

☐	0.50
☐	0.23
☐	0.127
☐	0.125



Question # 8 of 10 (Start time: 10:58:44 PM, 06 December 2023)

What is the bias of product estimator in two stage sampling?

Select the correct option



$$\text{Bias}(\bar{y}_{p2s}) = \bar{Y}(V_{01})$$



$$\text{Bias}(\bar{y}_{p2s}) = \bar{Y}(V_{11})$$



$$\text{Bias}(\bar{y}_{p2s}) = \bar{X}(V_{01})$$



None of the above

FastStone Capture



Question # 10 of 10 (**Start time: 11:00:54 PM, 06 December 2023**)

What is the formula of product estimator?

Select the correct option

☐	None of the above
☐	$\frac{\bar{X}\bar{y}}{\bar{x}}$
☒	$\frac{\bar{y}\bar{x}}{\bar{X}}$
☐	$\frac{\bar{X}\bar{x}}{\bar{y}}$



Question # 10 of 10 (Start time: 11:37:47 PM, 06 December 2023)

Total M

What is the R command for writing variance-covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.95 \\ 0.95 & 1 \end{bmatrix}$$

?

Select the correct option



Reload Math Equation

`sig=matrix(c(1,0.95,0.95,1),ncol=2)``sig=matrix(c(1,0.95,0.95,1),ncol=4)`

None of the above

`sig=matrix(c(1,1,0.65,0.65),ncol=2);`



BC230425602: MUHAMMAD HAMZA

Time Left 87
sec(s)

STA632 - Sampling Techniques (Quiz 2)

Quiz Start Time: 11:35 PM, 06 December 2023

Question # 9 of 10 (Start time: 11:37:40 PM, 06 December 2023)

Total Marks:

To develop interval estimate of any parameter of population, value which is added or subtracted from point estimate is classified as:

Select the correct option



margin of biasedness



margin of error



margin of efficiency



margin of consistency

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 11:37:18 PM, 06 December 2023)

Total Marks: 1

What is the bias of ratio estimator in two stage sampling?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--|
| ☐ | None of the above |
| ☒ | $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$ |
| ☐ | $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} - V_{20} - 2V_{11})$ |
| ☐ | $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} - V_{20} + 2V_{11})$ |

BC230425602: MUHAMMAD HAMZA

Time Left 79 sec(s)

STA632 – Sampling Techniques (Quiz 2)

Quiz Start Time: 11:35 PM, 06 December 2023

Question # 7 of 10 (Start time: 11:37:02 PM, 06 December 2023)

Total Marks:

Which is the formula of separate type ratio estimator (using 5 strata)?

Select the correct option

 Reload Math Equations

☐	$\bar{y}_{rsst} = \frac{1}{n_h} \sum_{h=1}^5 W_h \bar{y}_{rh}$
☒	$\bar{y}_{rsst} = \sum_{h=1}^5 W_h \bar{y}_{rh}$
☐	$\bar{y}_{rsst} = \sum_{h=1}^3 W_h \bar{y}_{rh}$
☐	$\bar{y}_{rsst} = \frac{1}{N_h} \sum_{h=1}^3 W_h \bar{y}_{rh}$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 11:36:48 PM, 06 December 2023)

Total Marks:

If $N = 2000$, $N_1 = 1000$; $n_1 = 150$, $N_2 = 500$; $N_3 = 500$; $n_2 = 8$; $n_3 = 25$, the value of W_1 is:

Select the correct option



0.50



0.23



0.51



0.125

BC230425602: MUHAMMAD HAMZA

Time Left 83 sec(s)

STA632 – Sampling Techniques (Quiz 2)

Quiz Start Time: 11:35 PM, 06 December 2023

Question # 5 of 10 (Start time: 11:36:36 PM, 06 December 2023)

Total Marks: 1

Which is the formula of separate type ratio estimator (using 8 strata)?

Select the correct option

 Reload Math Equations

☐

$$\bar{y}_{rsst} = \frac{1}{n_h} \sum_{h=1}^8 W_h \bar{y}_{rh}$$

☒

$$\bar{y}_{rsst} = \sum_{h=1}^8 W_h \bar{y}_{rh}$$

☐

$$\bar{y}_{rsst} = \frac{1}{N_h} \sum_{h=1}^5 W_h \bar{y}_{rh}$$

☐

$$\bar{y}_{rsst} = \sum_{h=1}^3 W_h \bar{y}_{rh}$$

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 11:36:02 PM, 06 December 2023)

What is MSE of product estimator in double sampling?

Select the correct option



Reload Math E



$$MSE(\bar{y}_{rd}) = \bar{Y}^2 [\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 + 2C_x C_y \rho xy)]$$



$$MSE(\bar{y}_{rd}) = \bar{Y}^2 [\lambda_2 C_x^2 + (\lambda_2 - \lambda_1)(C_y^2 + 2C_x C_y \rho xy)]$$



$$MSE(\bar{y}_{rd}) = \bar{X}^2 [\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 + 2C_x C_y \rho xy)]$$



None of the above

Click to Save Answer & Move to Next Q

Question # 3 of 10 (Start time: 11:35:46 PM, 06 December 2023)

Total Marks

Mrs. Trahan samples her class by selecting 5 girls and 7 boys. This type of sampling is called?

Select the correct option

- ☐ Simple
- ☐ Stratified
- ☒ Cluster
- ☐ Systematic

Not
same

[Click to Save Answer & Move to Next Question](#)

Question # 5 of 10 (Start time: 11:15:05 PM, 06 December 2023)

Total Marks

If the cluster sizes are unequal, the mean of cluster size is equal to

$$\bar{M} = \frac{M'}{N}$$

Select the correct option

 Reload Math Equations

☒	False
☐	True

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 11:13:39 PM, 06 December 2023)

Quiz Start Time: 11:12 PM, 06 D

If $N = 2000$, $N_1 = 1000$; $n_1 = 150$, $N_2 = 500$; $N_3 = 500$; $n_2 = 8$; $n_3 = 25$, the value of W is

Select the correct option

- | | |
|----------------------------------|-------|
| ☐ | 0.23 |
| ☒ | 0.50 |
| ☐ | 0.51 |
| ☐ | 0.125 |

Question # 1 of 10 (Start time: 11:12:03 PM, 06 December 2023)

Total Marks: 1

If $N = 6000$, $N_1 = 3000$; $n_1 = 150$, $N_2 = 2000$; $N_3 = 1000$; $n_2 = 8$; $n_3 = 25$, the value of W_3 is:

Select the correct option

☒	0.167
☐	0.40
☐	0.67
☐	0.132

spring
(2023)

Mid term

STA 632

(paper - 1)

- 1/ Find standard error of mean of following population
nechy values di gae thi 3,
- 2/ Define R code for showing 3 strata of following population 3 marks
- 3/ Define following cluster (as variable) in R code?
- 4/ Write R command to find the confidence interval for mean
clusters dia howy thy 5, marks
- 5/ Write R command to perform stratified sampling by Neyman allocation of sample size 14. 5, marks.

(paper - 2) By Queen Sahi :-

- 1/ Find n_1, n_2, p_1 and p_2 using populational allocation 3, marks
- 2/ Define sampling & sampling error. 5 marks.
- 3/ Ak table wala tha.
- 4/ prove ~~cluster~~ and ~~stratified~~ stratified.